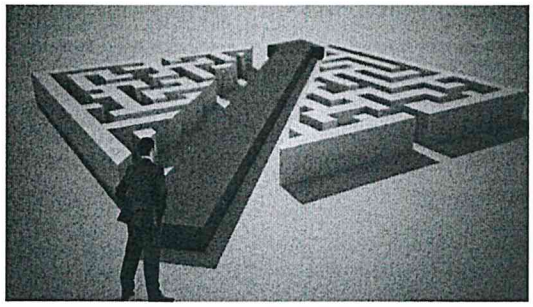
Minimum Distance from a Point to a Line



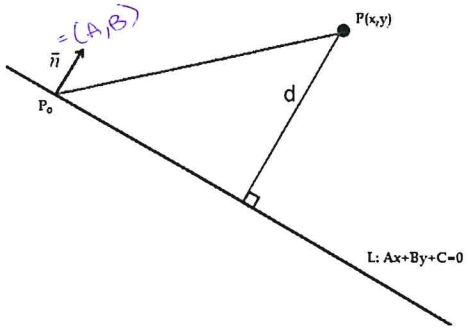


The shortest distance from a point to a line is the PERPENDICULAR distance.

Minimum distance in R^2

Recall: the magnitude of the vector projection of $\vec{a}$ onto $\vec{b}$ is $\dfrac{|\vec{a} \bullet \vec{b}|}{|\vec{b}|}$ ← magnitude / absolute value to ensure positive value for magnitude / length.

To determine the minimum distance from a point $P(x_1, y_1)$ to the line $Ax + By + C = 0$, let $P_0(x_0, y_0)$ be a known point on the line and let d be the minimum distance.



d = magnitude of projection of $\overrightarrow{P_0P}$ onto $\vec{n}$.

$$= \frac{|\overrightarrow{P_0P} \cdot \vec{n}|}{|\vec{n}|}$$

$$= \frac{|(x_1 - x_0, y_1 - y_0) \cdot (A, B)|}{\sqrt{A^2 + B^2}}$$

$$= \frac{|A(x_1 - x_0) + B(y_1 - y_0)|}{\sqrt{A^2 + B^2}}$$

$$= \frac{|Ax_1 + By_1 + (-Ax_0 - By_0)|}{\sqrt{A^2 + B^2}}$$

since (x_0, y_0) lies on the line.
$\therefore Ax_0 + By_0 + C = 0$
$\therefore C = -Ax_0 - By_0$

(x_1, y_1) is the point in R^2

$$\boxed{d = \frac{|Ax_1 + By_1 + C|}{\sqrt{A^2 + B^2}}}$$

Example 1: Determine the minimum distance from (6,-1) to the line $4x + y - 10 = 0$

$\vec{n} = (4, 1)$

$(x_1, y_1) = (6, -1)$

$$d = \frac{|Ax_1 + By_1 + C|}{\sqrt{A^2 + B^2}}$$

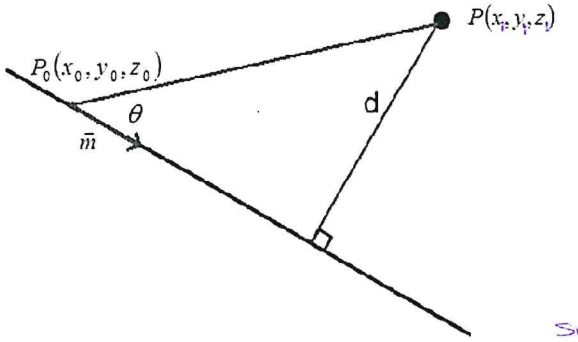
$$d = \frac{|4(6) + (-1) - 10|}{\sqrt{4^2 + 1^2}}$$

$$= \frac{|24 - 1 - 10|}{\sqrt{17}}$$

$$= \frac{|13|}{\sqrt{17}} = \frac{13\sqrt{17}}{17}$$

Minimum distance in $\mathbb{R}^3$

Recall: Lines in $\mathbb{R}^3$ do not have unique normal vectors.



Let $P(x, y, z)$ be a point in $\mathbb{R}^3$

Let $P_0(x_0, y_0, z_0)$ be a known point on the line with the direction vector $\vec{m}$

Let θ be the angle between $\overrightarrow{P_0P}$ and $\vec{m}$.

$$\sin\theta = \frac{d}{|\overrightarrow{P_0P}|}$$

$$d = |\overrightarrow{P_0P}| \sin\theta \times \frac{|\vec{m}|}{|\vec{m}|}$$

$$= \frac{|\overrightarrow{P_0P}|\,|\vec{m}|\sin\theta}{|\vec{m}|}$$

Recall: $|\vec{a} \times \vec{b}| = |\vec{a}||\vec{b}|\sin\theta$

$$\boxed{d = \frac{|\overrightarrow{P_0P} \times \vec{m}|}{|\vec{m}|}}$$

Example 2: Determine the minimum distance from (3,-2,1) to the line $\vec{r} = (1,0,-5) + t(2,-2,3),\ t \in \Re$.

$P(3,-2,1)$

$P_0(1,0,-5)$ ← known point on line $\Big\}$ $\overrightarrow{P_0P} = (2,-2,6)$

$\vec{m} = (2,-2,3)$

$$\overrightarrow{P_0P} \times \vec{m} = \begin{vmatrix} i & j & k \\ 2 & -2 & 6 \\ 2 & -2 & 3 \end{vmatrix} \begin{matrix} i & j \\ 2 & -2 \\ 2 & -2 \end{matrix}$$

$$= (-6+12,\ 12-6,\ -4+4)$$

$$= (6, 6, 0)$$

$$|\overrightarrow{P_0P} \times \vec{m}| = \sqrt{6^2+6^2+0^2}$$

$$= 6\sqrt{2}$$

$$|\vec{m}| = \sqrt{(2)^2+(-2)^2+(3)^2}$$

$$= \sqrt{17}$$

$$\therefore d = \frac{6\sqrt{2}}{\sqrt{17}}$$

$$d = \frac{6\sqrt{34}}{17}$$